

Homework 3

CENG 113, SP 2026

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Academic Integrity: For all problems **show your work** to receive credit. By submitting your assignment you're confirming that your submission is a product of your own effort.

Box all answers!

Problem 1

(Note: This is (4.B) from Homework 2. You're free to use the published solutions as a starting point.)

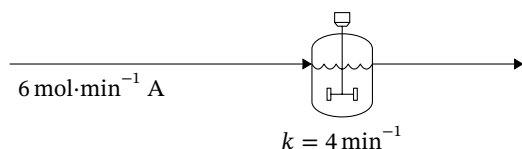
Ethylene oxide (A) is reacted with water (B) to form ethylene glycol (C, aka 1,2-ethanediol) according to the liquid-phase reaction $A + B \longrightarrow C$ which follows a **non-elementary**, first-order rate expression $r = kC_A$ with $k = 0.03 \text{ min}^{-1}$ at 50°C .

A solution containing 10 mol% ethylene oxide in water at 50°C is provided to an isothermal, 0.75 m^3 CSTR at a rate of $4 \text{ L}\cdot\text{s}^{-1}$. If necessary, the densities of all components can be assumed equal to the density of water ($1 \text{ kg}\cdot\text{L}^{-1}$).

Suppose two utility fluids are available: liquid cooling water at 15°C or superheated steam at 200°C . Estimate the heat exchanger area A required maintain isothermal operation. The heat capacity and enthalpy of formation of ethylene oxide are $48 \text{ J}\cdot\text{mol}^{-1}\cdot^\circ\text{C}^{-1}$ and $-52.6 \text{ kJ}\cdot\text{mol}^{-1}$. (ANS: $A \approx 7 \text{ m}^2$)

Problem 2

Consider the following CSTR in which the elementary, liquid-phase reaction $A \longrightarrow 2B$ takes place:

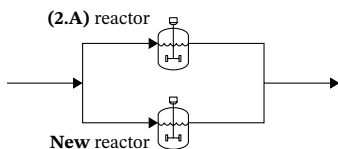


(2.A) Find the conversion of A if the residence time is 10 s. (ANS: 2/5)

(2.B) Another CSTR with the same volume as in the previous part is to be placed in series with the existing reactor (see below). This new reactor operates at a lower temperature so that the rate constant is reduced by 50% within the new reactor. Calculate the total outlet molar flow rate of A.



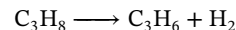
(2.C) The new CSTR (again at lower temperature such that the rate constant is again reduced by 50 as in (2.B)) could also be placed in parallel with the old reactor. Assuming n_{i0} and v_0 divide equally, calculate the total outlet molar flow rate of A. (ANS: about $3.0 \text{ mol}\cdot\text{min}^{-1}$)



(2.D) Calculate the outlet molar flow rate of B for the parallel and series arrangement. Which is preferred for production of B?

Problem 3

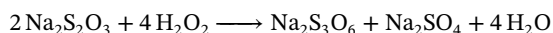
Gas-phase thermal degradation of propane (C_3H_8) to propylene (C_3H_6) and hydrogen (H_2) takes place in the presence of inert nitrogen (N_2) according to the elementary reaction



with $A = 0.6 \text{ s}^{-1}$ and $E_a = 14 \text{ kJ}\cdot\text{mol}^{-1}$. The feed is an equimolar mixture of propane and nitrogen at 375°C . Determine the conversion of propane achieved by an isothermal fluidized bed reactor if the residence time is 15 s. (ANS: $X_A \approx 30 - 40\%$)

Problem 4

The irreversible, aqueous reaction between sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$, which we'll call A) and hydrogen peroxide (H_2O_2 , which we'll call B) is *extremely* exothermic,



$$\Delta H \approx -1172 \text{ kJ}\cdot\text{mol}^{-1} \text{ (const)}$$

which makes it an interesting reaction for both kinetic and safety studies. It also happens to be a great reaction to study multiple steady states in stirred reactors.¹ Suppose we wish to study this reaction in a laboratory environment with a 500 mL, adiabatic CSTR. The feed contains 0.8 M each of $\text{Na}_2\text{S}_2\text{O}_3$ and H_2O_2 in deionized water and is provided so that the residence time τ is 50 s. Additional rate information is provided below.

$$r_A = -kC_A C_B \quad (\text{careful! This is } r_A, \text{ not } r!)$$

$$A = 2 \times 10^{10} \text{ L}\cdot\text{mol}^{-1}\cdot\text{s}^{-1}$$

$$E_a = 70 \text{ kJ}\cdot\text{mol}^{-1}$$

(4.A) Use Equation 5.8 and the quadratic formula (Equation A.1) to show that X_A can be written as

$$X_A = \frac{1 + 3\beta \pm \sqrt{\beta^2 + 6\beta + 1}}{4\beta}$$

where $\beta(T) = \tau C_{A0} k(T)$. Plot these two expressions for X_A on the same plot axes from 0°C to 100°C . Which solution is physically reasonable?

(4.B) Show that

$$G(T) = -\frac{1}{2} V k C_{A0}^2 \Delta H (1 - X_A)(1 - 2X_A)$$

(4.C) Assume the $\sum n_{i0} C_{p,i} (T - T_0)$ term in $R(T)$ can be approximated as though the inlet feed is pure water (therefore v_0 , n_{i0} , and $C_{p,i}$ can be treated as though they're for pure water for the purposes of calculating $R(T)$). In a new figure plot $G(T)$ as a red line and $R(T)$ as a blue line from 0°C to 100°C with $T_0 = 290 \text{ K}$. Put both curves on the same plot axes. Clearly label your axes with the appropriate units and include a legend.

(4.D) Estimate the extinction temperature and **explain** how you arrived at your estimate. What is the change in reactor temperature and conversion if the extinction process occurs? (hint: review Figs. 5.4 and 5.5) (ANS: Extinction is about $T_0 \approx 276 \text{ K}$)

¹ K.F. Lin; L.L. Wu. Performance of an adiabatic controlled cycled stirred tank reactor. Chem. Eng. Sci. 36 (1981) 435-444. S.A. Vejtasa; R.A. Schmitz. An experimental study of steady state multiplicity and stability in an adiabatic stirred reactor. AIChE Journal, 16 (1970) 410-419.